



Python Data Types

Topics:

- Basic Data Types in Python
- Handling Data in Python

Data types in Python

Data types are the classification or categorization of data items. It represents the kind of value that tells what operations can be performed on a particular data. Since everything is an object in Python programming, data types are classes and variables are instance (object) of these classes.

Note – `type()` function is used to determine the type of data type.

Numeric

In Python, numeric data type represent the data which has numeric value. Numeric value can be integer, floating number or even complex numbers. These values are defined as int, float and complex class in Python.

Integers – This value is represented by int class. It contains positive or negative whole numbers (without fraction or decimal). In Python there is no limit to how long an integer value can be.

Float – This value is represented by float class. It is a real number with floating point representation. It is specified by a decimal point. Optionally, the character e or E followed by a positive or negative integer may be appended to specify scientific notation.

Complex Numbers – Complex number is represented by complex class. It is specified as *(real part) + (imaginary part)j*. For example – 2+3j

Example

```
>>> x = 100
```

```
>>> print(type(x))
```

```
<class 'int'>
```

```
>>> x = 10.25
```

```
>>> print(type(x))
```

```
<class 'float'>
```

```
>>> x = 3 + 4j
```

```
>>> print(type(x))
```

```
<class 'complex'>
```

String

In Python, Strings are arrays of bytes representing Unicode characters. A string is a collection of one or more characters put in a single quote, double-quote or triple quote. In python there is no character data type, a character is a string of length one. It is represented by `str` class.

Example

```
>>> s = "welcome to Python"
```

```
>>> print(s)
```

```
welcome to Python
```

```
>>> print(type(s))
```

```
<class 'str'>
```

```
>>> x = '''Welcome to Python'''
```

```
'Welcome to Python'
```

```
>>> print(x)
```

```
Welcome to Python
```

```
>>> print(type(x))
```

```
<class 'str'>
```

Boolean

Data type with one of the two built-in values, `True` or `False`. Boolean objects that are equal to `True` are *truthy* (`true`), and those equal to `False` are *falsy* (`false`). But non-Boolean objects can be evaluated in Boolean context as well and determined to be *true* or *false*. It is denoted by the class `bool`.

Note – `True` and `False` with capital 'T' and 'F' are valid Booleans otherwise python will throw an error.

Example

```
>>> x = True
```

```
>>> print(x)
```

```
True
```

```
>>> print(type(x))
```

```
<class 'bool'>
```

```
>>> y = False
```

```
>>> print(y)
```

```
False
```